

ASSIGNMENT - CHOWLY

All over the world, top class restaurants have looked at ways to have an edge over their competitors. Having extra ordinary chefs and aesthetically pleasing interiors are some key details these restaurants do not joke with. However, a couple of restaurants have stepped up their game to provide digital dining experience even for their physical customers. Chowly is an app solution adopted by restaurants which serves as a customer's guide.

Features;

A visiting customer can view the restaurant's food and drinks menus and place an order. The waiting time for each order can be viewed by the customer along with other order details. After a customer submits an order, the order is assigned to a waiter who updates the order details by specifying the chef and the bartender who prepared the order on the platform. In a case when there is serious delay in the preparation of the order, the customer can submit a complaint and give a low rating on that order. Most importantly, payment for the order is made on the platform just before the customer exits the restaurant.

Task

You have been hired as the Software Engineer for Chowly. Produce a well-engineered model for the solution.

CHOWLY – Engineered Model

Chowly is an app solution adopted by restaurants to serve as a customer's guide for digital dining. It allows customers to view food and drinks menus, place orders, track waiting times, submit complaints, rate orders, and make payments all from the platform. The following engineered model captures every entity, attribute, relationship, and sample record required to implement the Chowly system.

The model consists of 12 entities with their complete attributes, primary keys, foreign keys, and records.

Entities:

- RESTAURANT entity.
- MENU entity.
- MENUITEM entity.
- CUSTOMER entity.
- WAITER entity.
- CHEF entity.
- BARTENDER entity.
- ORDER entity.
- ORDERITEM entity.
- COMPLAINT entity.
- RATING entity.
- PAYMENT entity.

RESTAURANT

for keeping records of restaurants that adopt the Chowly platform

RestaurantID (PK)	RestaurantName	RestaurantLocation	RestaurantPhone	RestaurantEmail
R001	The Golden Gate	13, Ubah str , Berger	+234-801-001	info@goldengatecom
R002	Bangtan bistro	45 Allen Ave, Apapa	+234-201-002	hello@bangtanbistro.com
R003	Noiis Place	12, Forcados, Festac	+234-720-003	contact@noiisplace.co.ke

MENU

for keeping records of food and drinks menus offered by a restaurant

MenuID (PK)	MenuName	MenuType	MenuRestaurantId (FK)
M001	Main Course	Food	R001
M002	Cocktails & Drinks	Drinks	R001
M003	Breakfast Menu	Food	R002
M004	Wine List	Drinks	R002
M005	Seafood Specials	Food	R003

MENUITEM

for keeping records of individual food and drink items available on a menu

MenuItemID (PK)	MenuItemName	MenuItemDescription	MenuItemPrice	MenuItemCategory	MenuItemAvailability	MenuItemMenuId (FK)
MI001	Grilled Steak	Premium cut sirloin with herbs	8500	Food	Available	M001
MI002	Jollof Rice	Spicy tomato based rice	3500	Food	Available	M001
MI003	Mojito	Refreshing lime and mint cocktail	4000	Drinks	Available	M002
MI004	Chapman	Mixed fruit cocktail with bitters	3500	Drinks	Available	M002
MI005	Eggs Benedict	Poached eggs on English muffin	5000	Food	Available	M003
MI006	Merlot 2018	French red wine glass	6000	Drinks	Available	M004

CUSTOMER

for keeping records of visiting customers who use the Chowly app

CustomerID (PK)	CustomerName	CustomerPhone	CustomerEmail	CustomerTableNo
C001	Amina Yusuf	+234-803-111	amina@email.com	T5
C002	Kwame Asante	+234-501-222	kwame@email.com	T3
C003	Fatima Njoku	+234-805-333	fatima@email.com	T8

WAITER

for keeping records of waiters assigned to handle orders

WaiterID (PK)	WaiterName	WaiterPhone	WaiterShift	WaiterRestaurantId (FK)
W001	Daniel Osei	+234-806-111	Evening	R001
W002	Grace Mbeki	+233-502-222	Morning	R002
W003	Samuel Kariuki	+254-721-333	Evening	R003

CHEF

for keeping records of chefs who prepare food orders

ChefID (PK)	ChefName	ChefSpecialty	ChefExperience	ChefRestaurantId (FK)
CH001	Chef Adekunle	Continental Cuisine	10 yrs	R001

ChefID (PK)	ChefName	ChefSpecialty	ChefExperience	ChefRestaurantId (FK)
CH002	Chef Nkrumah	African Dishes	8 yrs	R002
CH003	Chef Wanjiku	Seafood	12 yrs	R003

BARTENDER

for keeping records of bartenders who prepare drink orders

BartenderID (PK)	BartenderName	BartenderSpecialty	BartenderShift	BartenderRestaurantId (FK)
B001	Tunde Bakare	Cocktails	Evening	R001
B002	Ama Mensah	Wine & Spirits	Morning	R002
B003	Juma Otieno	Mocktails & Juices	Evening	R003

ORDER

for keeping records of customer orders including waiting time and status

OrderID (PK)	OrderDate	OrderTime	OrderStatus	OrderWaitTime	OrderCustomerId (FK)	OrderWaiterId (FK)	OrderChefId (FK)	OrderBartenderId (FK)
O001	2026-08-20	19:30	Completed	25 mins	C001	W001	CH001	B001
O002	2026-08-20	12:15	Completed	40 mins	C002	W002	CH002	B002
O003	2026-08-21	20:00	Delayed	60 mins	C003	W003	CH003	B003

ORDERITEM

serves as a bridge table for the many-to-many relationship between ORDER and MENUITEM

OrderItemID (PK)	OrderItemQuantity	OrderItemSubtotal	OrderItemOrderId (FK)	OrderItemMenuItemId (FK)
OI001	1	8500	O001	MI001
OI002	1	3500	O001	MI002
OI003	2	8000	O001	MI003
OI004	1	5000	O002	MI005
OI005	1	6000	O002	MI006
OI006	1	3500	O003	MI004

COMPLAINT

for keeping records of complaints submitted by customers regarding delayed orders

ComplaintID (PK)	ComplaintDescription	ComplaintDate	ComplaintTime	ComplaintOrderId (FK)	ComplaintCustomerId (FK)
CMP001	Order took over 60 minutes, unacceptable delay	2026-08-21	21:05	O003	C003
CMP002	Wrong drink served, had to wait for replacement	2026-08-21	20:45	O003	C003

ComplaintID (PK)	ComplaintDescription	ComplaintDate	ComplaintTime	ComplaintOrderId (FK)	ComplaintCustomerId (FK)
CMP003	Food was cold upon arrival after long wait	2026-08-20	21:30	O002	C002

RATING

for keeping records of ratings given by customers on orders

RatingID (PK)	RatingScore	RatingComment	RatingDate	RatingOrderId (FK)	RatingCustomerId (FK)
RAT001	2	Very slow service, waited too long	2026-08-21	O003	C003
RAT002	4	Good food, reasonable wait	2026-08-20	O001	C001
RAT003	3	Average experience, drink was wrong initially	2026-08-20	O002	C002

PAYMENT

for keeping records of payments made by customers on the platform before exiting the restaurant

PaymentID (PK)	PaymentAmount	PaymentMethod	PaymentDate	PaymentTime	PaymentStatus	PaymentOrderId (FK)	PaymentCustomerId (FK)
PAY001	20000	Card	2026-08-20	21:15	Successful	O001	C001

PaymentID (PK)	PaymentAmount	PaymentMethod	PaymentDate	PaymentTime	PaymentStatus	PaymentOrderId (FK)	PaymentCustomerId (FK)
PAY002	11000	Mobile Money	2026-08-20	13:30	Successful	O002	C002
PAY003	3500	Card	2026-08-21	22:00	Successful	O003	C003

Relationships Summary (Foreign Keys)

The following table summarises every foreign key in the model, the relationship it creates, and the cardinality between the parent and child entities.

FK Column	Child Entity	Parent Entity	Relationship	Relationship
MenuRestaurantId	MENU	RESTAURANT	Restaurant has Menus	1:M
MenuItemMenuId	MENUITEM	MENU	Menu has MenuItems	1:M
WaiterRestaurantId	WAITER	RESTAURANT	Restaurant has Waiters	1:M
ChefRestaurantId	CHEF	RESTAURANT	Restaurant has Chefs	1:M
BartenderRestaurantId	BARTENDER	RESTAURANT	Restaurant has Bartenders	1:M
OrderCustomerId	ORDER	CUSTOMER	Customer places Orders	1:M
OrderWaiterId	ORDER	WAITER	Waiter handles Orders	1:M
OrderChefId	ORDER	CHEF	Chef prepares Orders	1:M
OrderBartenderId	ORDER	BARTENDER	Bartender prepares Orders	1:M
OrderItemOrderId	ORDERITEM	ORDER	Order contains OrderItems	1:M

FK Column	Child Entity	Parent Entity	Relationship	Relationship
OrderItemMenuItemId	ORDERITEM	MENUITEM	Menuitem appears in OrderItems	1:M
ComplaintOrderId	COMPLAINT	ORDER	Order receives Complaints	1:M
ComplaintCustomerId	COMPLAINT	CUSTOMER	Customer submits Complaints	1:M
RatingOrderId	RATING	ORDER	Order receives Ratings	1:M
RatingCustomerId	RATING	CUSTOMER	Customer gives Ratings	1:M
PaymentOrderId	PAYMENT	ORDER	Order has Payment	1:1
PaymentCustomerId	PAYMENT	CUSTOMER	Customer makes Payments	1:M